Similarity & Diagonalization

Similarity

Theorem

The next theorem illustrates one use of the characteristic polynomial, and it provides the foundation for several iterative methods that *approximate* eigenvalues. If A and B are $n \times n$ matrices, then A **is similar to** B if there is an invertible matrix P such that $P^{-1}AP = B$, or, equivalently, $A = PBP^{-1}$. Writing Q for P^{-1}, we have $Q^{-1}BQ = A$. So B is also similar to A, and we say simply that A and B **are similar**. Changing A into $P^{-1}AP$ is called a **similarity transformation**.

If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

In linear algebra, the concept of similarity, where two matrices represent the same linear transformation under different bases, is studied to simplify complex matrix analysis and solve various problems. Specifically, it's crucial for understanding how transformations change with coordinate system changes and for diagonalizing matrices, which facilitates easier analysis and solution of systems of equations

Two square matrices are said to be similar if they represent the same linear operator under different bases.
Two similar matrices have the same rank, trace, determinant, and eigenvalues.

When we say two **square matrices are similar**, we mean they represent the **same linear transformation but** expressed in **different coordinate systems or bases**.

A **basis** for a vector space is a set of linearly independent vectors that **span the space**. This means:

•Any vector in the space can be written as a **unique combination** of the basis vectors.
•For example, in 2D space, the standard basis is:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

what's the meaning of different bases?
It means we are **looking at the same transformation but** describing it **from different perspectives** (like changing camera angles).

Analogy:
Imagine rotating a 3D object. If you observe it from two different angles, the **object doesn’t change**, but your **description** of its coordinates does.

We begin this section by recalling the definition of similar matrices. Recall that if A,B are two $n\times n$ matrices, then they are similar if and only if there exists an invertible matrix P such that

$$A = P^{-1}BP$$

Here, A and B represent the same linear transformation.
But:
A describes the transformation in one basis.
B describes it on another basis.
P is the change of basis matrix.

Similarity is an equivalence relation, i.e. for $n \times n$ matrices A, B, and C,

1. $A \sim A$ (reflexive)
2. If $A \sim B$, then $B \sim A$ (symmetric)
3. If $A \sim B$ and $B \sim C$, then $A \sim C$ (transitive)

Let A, B, and C be $n \times n$ matrices.

1. **Reflexivity:** A is similar to itself.
2. **Symmetry:** if A is similar to B, then B is similar to A.
3. **Transitivity:** if A is similar to B and B is similar to C, then A is similar to C.

Why is similarity important?
Since similar matrices represent the same transformation in different bases, they must share **invariant properties**, such as:

- **Rank**
- **Trace**
- **Determinant**
- **Eigenvalues**

These are things that do **not change** even when you change the basis.

The ability to diagonalize matrices through similarity transformations is valuable in various fields, including differential equations, systems of equations, and more.

It is clear that $A \sim A$, taking $P = I$.

Now, if $A \sim B$, then for some P invertible,

$$A = P^{-1}BP$$

and so

$$PAP^{-1} = B$$

But then

$$\left(P^{-1}\right)^{-1}AP^{-1} = B$$

which shows that $B \sim A$.

Now suppose $A \sim B$ and $B \sim C$. Then there exist invertible matrices P, Q such that

$$A = P^{-1}BP,\ B = Q^{-1}CQ$$

Then,

$$A = P^{-1}\left(Q^{-1}CQ\right)P = (QP)^{-1}C\,(QP)$$

showing that A is similar to C.

Key Properties of Similarity:

1. **Preservation of Eigenvalues:** Similar matrices have the same eigenvalues (including their algebraic multiplicities). This is a fundamental property that makes similarity important in many applications.
2. **Same Determinant**: If A and B are similar, then $\det(A) = \det(B)$.
3. **Same Trace:** Similar matrices have the same trace ($\mathrm{tr}(A) = \mathrm{tr}(B)$), which is the sum of their eigenvalues.
4. **Equivalent Characteristic Polynomials**: A and B have the same characteristic polynomial.
5. **Diagonalizability**: If a matrix A is diagonalizable, it is similar to a diagonal matrix. That is, there exists an invertible matrix P such that:

   $$A = PDP^{-1}$$

   where D is diagonal.

6. **Relation to Basis Change**: Similarity represents a change of basis. If A is the matrix representation of a linear transformation in one basis, B represents the same transformation in another basis.

Applications:

•**Simplifying Matrices**: Similarity transformations are often used to simplify computations, such as reducing a matrix to Jordan form or diagonal form.
•**Control Theory**: In engineering, similarity transformations help to analyze and design systems.
•**Quantum Mechanics**: Hermitian similarity (a specific kind of similarity) is used in quantum mechanics to study operators.

In the context of matrix similarity, the matrix P is an **invertible matrix** that facilitates the transformation between the two similar matrices A and B. Here's what P represents:

1. **Change of Basis Matrix**:
   - P represents a change of basis in the vector space associated with the matrices A and B.
   - A is the representation of a linear transformation in one basis, and B is the representation of the same transformation in a different basis.
2. **How It Works:**
   - The similarity transformation $B = P^{-1}AP$ modifies A by "re-expressing" it in a new basis, defined by P.
   - The columns of P are typically the basis vectors of the new basis in which the transformation is represented by B.
3. **Properties of P**:
   - P must be **invertible** (i.e., it has a nonzero determinant). If P is not invertible, the transformation is not well-defined.
   - P^{-1} is the inverse of P, satisfying $PP^{-1} = I$, where I is the identity matrix.

Suppose:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Then $B = P^{-1}AP$ is computed as:

1. Compute P^{-1}:

$$P^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

2. Compute B:

$$B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

After performing the matrix multiplications, B is a new matrix similar to A.

Choosing the matrix P in a similarity transformation $B = P^{-1}AP$ depends on the context or purpose of the transformation.

- Compute the eigenvalues of A.
- Find the eigenvectors corresponding to each eigenvalue.
- Form P by arranging these eigenvectors as columns.

General Criteria for P:

Invertibility: P must be invertible.

Application-Dependent: The choice of P depends on whether the goal is diagonalization, Jordan form, or simplifying A for analysis.

Let's consider the matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

We will find a matrix P and compute $B = P^{-1}AP$.

Let's choose:

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Now compute P^{-1}:

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Now compute $B = P^{-1}AP$:

$$B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Step-by-step:

1. Compute AP:

$$AP = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$$

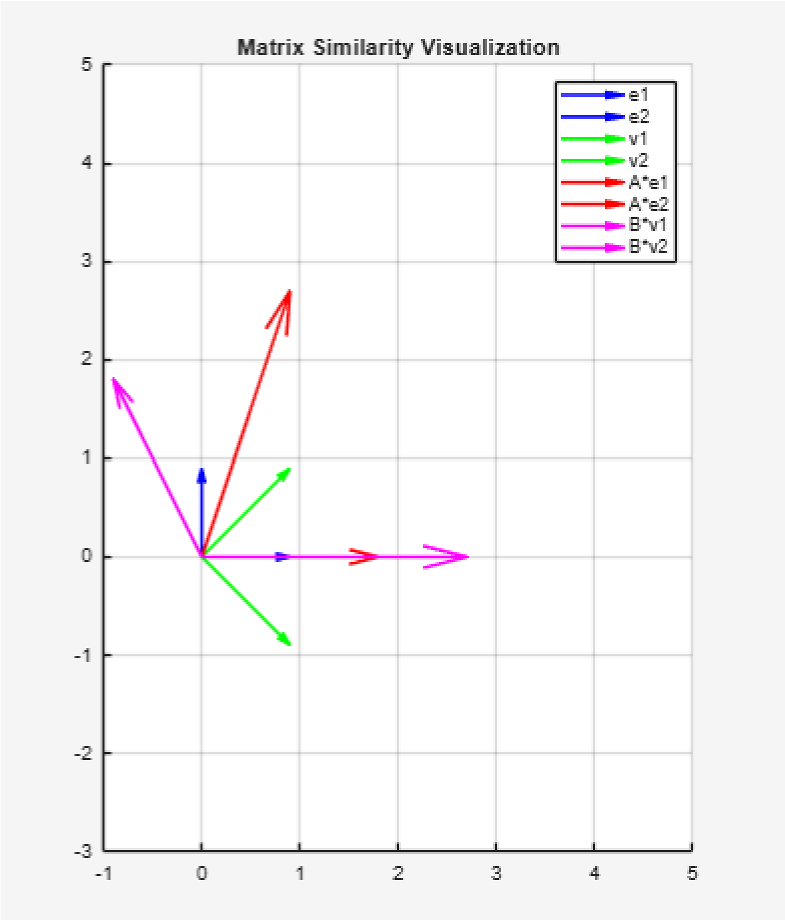
2. Compute $P^{-1}(AP)$:

$$B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

So $B = A$, which confirms that similarity is preserved under change of basis.

Key Properties

- Trace: $\mathrm{tr}(A) = 4 = \mathrm{tr}(B)$
- Determinant: $\det(A) = 4 = \det(B)$
- Eigenvalues: Both have eigenvalue 2 (with multiplicity 2)
- Rank: Both have rank 2



Matrix Similarity Visualization
e1
e2
v1
v2
A*e1
A*e2
B*v1
B*v2
5
4
3
2
1
0
-1
-2
-3
-1
0
1
2
3
4
5

Diagonalization

In many cases, the eigenvalue–eigenvector information contained within a matrix A can be displayed in a useful factorization of the form $A = PDP^{-1}$ where D is a diagonal matrix. In this section, the factorization enables us to compute A^k quickly for large values of k, a fundamental idea in several applications of linear algebra.

A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D.

Note If $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$, then $D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$

$$D^3 = DD^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

In general, $D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix}$ for $k \geq 1$

We study **diagonalization in linear algebra** because it gives us a **powerful and simplified way to understand and compute with linear transformations**.

Diagonalization means rewriting a matrix in a simpler form — a **diagonal matrix** — using a **change of basis**

Simplifies Matrix Powers

If A is diagonalizable, then:

$$A = PDP^{-1}$$

where D is diagonal and P is the matrix of eigenvectors.

Then:

$$A^n = (PDP^{-1})^n = PD^nP^{-1}$$

And since taking powers of a diagonal matrix is **very easy** (just raise each diagonal element to the power), this is super efficient.

Example:

If $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$,

then $D^5 = \begin{bmatrix} 2^5 & 0 \\ 0 & 3^5 \end{bmatrix}$

Solving Systems of Differential Equations

Many systems of linear differential equations use matrix exponentials:

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

Solving this becomes easier if A is diagonalizable:

$$e^{At} = Pe^{Dt}P^{-1}$$

Again, exponentiating diagonal matrices is very simple.

Understanding Linear Transformations

Diagonalization gives deep insight into what a transformation is doing:

- The diagonal entries in D are the **eigenvalues**.
- The columns of P are the **eigenvectors**.

This tells you **how the transformation stretches or compresses** space in specific directions.

- **Quantum mechanics** (observables are represented by diagonalizable matrices).
- **PCA (Principal Component Analysis)** in data science (eigenvectors of covariance matrices).
- **Markov chains** and **Google's PageRank**.
- **Control systems**, **vibrations**, and more.

We study diagonalization because it:
Simplifies computations
Reveals the structure of a transformation
It is essential in many applications across science and engineering

Let A be an $n \times n$ matrix. Then A is said to be **diagonalizable** if there exists an invertible matrix P such that

$$P^{-1}AP = D$$

where D is a diagonal matrix.

An $n \times n$ matrix A is diagonalizable if and only if there is an invertible matrix P given by

$$P = \begin{bmatrix} X_1 & X_2 & \cdots & X_n \end{bmatrix}$$

where the X_k are eigenvectors of A.

Moreover, if A is diagonalizable, the corresponding eigenvalues of A are the diagonal entries of the diagonal matrix D.

Suppose P is given as above as an invertible matrix whose columns are eigenvectors of A. Then P^{-1} is of the form

$$P^{-1} = \begin{bmatrix} W_1^T \\ W_2^T \\ \vdots \\ W_n^T \end{bmatrix}$$

where $W_k^T X_j = \delta_{kj}$, which is the Kronecker's symbol defined by

$$\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

Then

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} W_1^T \\ W_2^T \\ \vdots \\ W_n^T \end{bmatrix} [\, AX_1 \quad AX_2 \quad \cdots \quad AX_n \,] \\ &= \begin{bmatrix} W_1^T \\ W_2^T \\ \vdots \\ W_n^T \end{bmatrix} [\, \lambda_1 X_1 \quad \lambda_2 X_2 \quad \cdots \quad \lambda_n X_n \,] \\ &= \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \end{aligned}$$

Conversely, suppose A is diagonalizable so that $P^{-1}AP = D$. Let

$$P = [\,X_1 \quad X_2 \quad \cdots \quad X_n\,]$$

where the columns are the X_k and

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

Then

$$AP = PD = [\,X_1 \quad X_2 \quad \cdots \quad X_n\,] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

and so

$$[\,AX_1 \quad AX_2 \quad \cdots \quad AX_n\,] = [\,\lambda_1 X_1 \quad \lambda_2 X_2 \quad \cdots \quad \lambda_n X_n\,]$$

showing the X_k are eigenvectors of A and the λ_k are eigenvectors.

Diagonalization

Four Step Method

Step 1. *Find the eigenvalues of A.*

Step 2. *Find three linearly independent eigenvectors of A.*

Step 3. *Construct P from the vectors in step 2.*

Step 4. *Construct D from the corresponding eigenvalues.*

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1: Find the eigenvalues

We solve the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

So:

$$\det\left(\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}\right) = (4-\lambda)(3-\lambda) - (2)(1)$$

$$= (4-\lambda)(3-\lambda) - 2 = (12 - 4\lambda - 3\lambda + \lambda^2) - 2 = \lambda^2 - 7\lambda + 10$$

Solve:

$$\lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$$

So the eigenvalues are:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

Step 2: Find the eigenvectors

For $\lambda = 5$:

Solve $(A - 5I)\mathbf{v} = 0$:

$$A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Row reduce or observe:

$$-1x + 1y = 0 \Rightarrow x = y$$

So eigenvector is any multiple of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 2$:

$$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \Rightarrow 2x + y = 0 \Rightarrow y = -2x$$

So eigenvector is any multiple of $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Step 3: Form matrices P and D

Use the eigenvectors as **columns of** P:

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

So we have:

$$A = PDP^{-1}$$

You can check this by computing P^{-1}, then confirming $PDP^{-1} = A$

Summary

- $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$
- Diagonalizable: YES
- Eigenvalues: 5 and 2
- Eigenvectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}$
- Diagonal form: $$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$